

Deep Learning

Theoretical Exercises - Lesson 4 - Chapter 5

Exercises on the book "Deep Learning" written by Ian Goodfellow,
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1 Exercises on Machine Learning Basics

**Hint:**

Several answers are correct in the multiple choice exercises.

1. The goal of machine learning is to achieve ...

☐ ... a small training error.

☐ ... a large training error.

☒ ... a small test error.

☐ ... a large test error.

☒ ... a small generalization error.

The test error is an estimation of the generalization error.

☐ ... a large generalization error.

2. An overfitted model has ...

☒ ... a large test error.

☐ ... a small test error.

☐ ... a large training error.

☒ ... a small training error.

3. An underfitted model has ...

☒ ... a large test error.

☐ ... a small test error.

☒ ... a large training error.

☐ ... a small training error.

4. A model tends to overfit when ...

- ☒ ... the training set is small.
- ☒ ... the regularization term has little weight.
- ☐ ... the capacity is smaller than the complexity of the task.
- ☐ ... the test error is close to the Bayes error.
- ☒ ... the training error is smaller than the Bayes error.

5. To prevent overfitting one can ...

- ☐ ... use a smaller test set.
- ☐ ... use a larger test set.
- ☐ ... use a smaller training set.
- ☒ ... use a larger training set.
- ☒ ... reduce the capacity of the model.
- ☐ ... increase the capacity of the model.

6. To prevent underfitting one can ...

- ☐ ... use a smaller test set.
- ☐ ... use a larger test set.
- ☐ ... use a smaller training set.
- ☐ ... use a larger training set.
- ☐ ... reduce the capacity of the model.
- ☒ ... increase the capacity of the model.

7. The goal of regularization is to reduce ...

- ☐ ... the training error.
- ☒ ... the generalization error.
- ☒ ... the test error.
- ☐ ... the Bayes error.

8. Mark the correct statements and correct the wrong ones.

- ☒ The test set is used to estimate the generalization error.
- ☐ The ~~training~~ **validation** set is used to control the training.
- ☐ The ~~validation~~ **training** set is used to learn the task.
- ☐ The training error typically underestimates the generalization error by a ~~smaller~~ **larger** amount than the validation error.
- ☒ The validation set is used to learn the hyperparameters.

9. Given is a set of samples $\{x^{(1)}, \dots, x^{(m)}\}$ that are independently and identically distributed according to a uniform distribution on the interval $[-0.8, 1.2]$.

- (a) Check whether the sample mean

$$\hat{\mu}_m = \frac{1}{m} \sum_{i=1}^m x^{(i)} \quad (1.1)$$

is an unbiased estimator of the true mean μ .

- (b) Assume that the absolute value of each sample is accidentally taken before the sample mean value is calculated. Thus the new estimator is

$$\hat{\mu}_m = \frac{1}{m} \sum_{i=1}^m |x^{(i)}|. \quad (1.2)$$

Determine the bias of this poor estimator to the mean μ of the initial distribution.

- (c) How can the estimator of (b) be fixed so that he still gives an unbiased estimate?

Solution:

- (a) The samples are distributed according to the distribution

$$p(x^{(i)}) = \begin{cases} \frac{1}{2}, & -0.8 \leq x^{(i)} \leq 1.2 \\ 0, & \text{otherwise} \end{cases},$$

which results in the true mean value of

$$\mu = \frac{a+b}{2} = \frac{(-0.8) + 1.2}{2} = 0.2.$$

The expected value of the sample mean (1.1) is

$$\begin{aligned} \mathbb{E}[\hat{\mu}_m] &= \mathbb{E}\left[\frac{1}{m} \sum_{i=1}^m x^{(i)}\right] \\ &= \frac{1}{m} \sum_{i=1}^m \mathbb{E}[x^{(i)}] \\ &= \frac{1}{m} \sum_{i=1}^m \left(\int_{-0.8}^{1.2} x^{(i)} \frac{1}{2} dx^{(i)} \right) \\ &= \frac{1}{m} \sum_{i=1}^m 0.2 \\ &= 0.2, \end{aligned}$$

resulting in a bias of 0:

$$\begin{aligned} \text{bias}(\hat{\mu}_m) &= \mathbb{E}[\hat{\mu}_m] - \mu \\ &= 0.2 - 0.2 \\ &= 0. \end{aligned}$$

Thus, the sample mean (1.1) is an unbiased estimator.

- (b) To determine the bias of the modified sample mean (1.2), its expected value has to be calculated as follows:

$$\begin{aligned}
\mathbb{E}[\hat{\mu}_m] &= \mathbb{E}\left[\frac{1}{m} \sum_{i=1}^m |x^{(i)}|\right] \\
&= \frac{1}{m} \sum_{i=1}^m \mathbb{E}[|x^{(i)}|] \\
&= \frac{1}{m} \sum_{i=1}^m \left(\int_{-0.8}^{1.2} |x^{(i)}| \frac{1}{2} dx^{(i)} \right) \\
&= \frac{1}{m} \sum_{i=1}^m \left(\int_{-0.8}^0 |x^{(i)}| \frac{1}{2} dx^{(i)} + \int_0^{1.2} |x^{(i)}| \frac{1}{2} dx^{(i)} \right) \\
&= \frac{1}{m} \sum_{i=1}^m \left(\int_{-0.8}^0 -x^{(i)} \frac{1}{2} dx^{(i)} + \int_0^{1.2} x^{(i)} \frac{1}{2} dx^{(i)} \right) \\
&= \frac{1}{m} \sum_{i=1}^m (0.16 + 0.36) \\
&= \frac{1}{m} \sum_{i=1}^m 0.52 \\
&= 0.52.
\end{aligned}$$

This results in a bias of

$$\begin{aligned}
\text{bias}(\hat{\mu}_m) &= \mathbb{E}[\hat{\mu}_m] - \mu \\
&= 0.52 - 0.2 \\
&= 0.32.
\end{aligned}$$

Thus, the modified sample mean (1.2) clearly is an biased estimator.

- (c) To fix this biased estimator, one can either just subtract the bias term

$$\begin{aligned}
\hat{\mu}_m &= \frac{1}{m} \sum_{i=1}^m |x^{(i)}| - \text{bias}(\hat{\mu}_m) \\
&= \frac{1}{m} \sum_{i=1}^m |x^{(i)}| - 0.32
\end{aligned}$$

or add a large number C to each sample, which must be subtracted again after the mean value has been calculated:

$$\hat{\mu}_m = \frac{1}{m} \sum_{i=1}^m |x^{(i)} + C| - C.$$

This number must be greater than the absolute value of the lower limit of the interval ($C > |a|$).

The second approach is simpler, since the bias does not have to be calculated.

10. **Optional:** Consider a set of samples $\{x^{(1)}, \dots, x^{(m)}\}$ that are independently and identically distributed according to a uniform distribution on the interval $[0, \theta]$, thus

$$p(x^{(i)}, \theta) = \begin{cases} \frac{1}{\theta}, & 0 \leq x^{(i)} \leq \theta \\ 0, & \text{otherwise} \end{cases}.$$

A biased estimator for the parameter θ is

$$\hat{\theta} = \max(x^{(1)}, \dots, x^{(m)}).$$

Correct this estimator so that it becomes an unbiased estimator for θ .

Solution:

In order to calculate the expected value $\mathbb{E}[\hat{\theta}]$, the probability density function must first be calculated from the distribution function of $\hat{\theta}$. The distribution function of X on the interval $[0, \theta]$ is

$$F(x) = P(X \leq x) = \frac{x}{\theta}.$$

Hence, the distribution function of $\hat{\theta} = \max(x^{(1)}, \dots, x^{(m)})$ is

$$\begin{aligned} P(\hat{\theta} \leq x) &= P(\max(x^{(1)}, \dots, x^{(m)}) \leq x) \\ &= P(x^{(1)} \leq x \cap \dots \cap x^{(m)} \leq x) \\ &= P(x^{(1)} \leq x) \cdot \dots \cdot P(x^{(m)} \leq x) \\ &= \frac{x}{\theta} \cdot \dots \cdot \frac{x}{\theta} \\ &= \frac{x^m}{\theta^m}. \end{aligned}$$

By derivating this function, the probability density function can be calculated as:

$$p_{\hat{\theta}}(x) = \frac{m x^{m-1}}{\theta^m}.$$

With this it is now possible to calculate the expected value

$$\begin{aligned} \mathbb{E}[\hat{\theta}] &= \int_0^\theta x p_{\hat{\theta}}(x) dx \\ &= \int_0^\theta x \frac{m x^{m-1}}{\theta^m} dx \\ &= \frac{m}{\theta^m} \int_0^\theta x^m dx \\ &= \frac{m}{\theta^m} \left[\frac{x^{m+1}}{m+1} \right]_0^\theta \\ &= \frac{m}{\theta^m} \cdot \frac{\theta^{m+1}}{m+1} \\ &= \frac{m}{m+1} \theta. \end{aligned}$$

Except for factor $\frac{m}{m+1}$, this corresponds exactly to the required interval length θ . Hence, an unbiased estimator for the parameter θ is

$$\hat{\theta} = \frac{m+1}{m} \max(x^{(1)}, \dots, x^{(m)}).$$